

STAT 516: Homework 6

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(15 points)** Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) **(8 points)** Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
- (b) **(7 points)** Determine whether X and Y are independent.

2. **(13 points)** Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5}(x^2 + y), & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[2X - Y + XY].$$

3. **(12 points)** Let X represent the age of an insured automobile involved in an accident. Let Y represent the length of time the owner has insured the automobile at the time of the accident. Suppose X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{64}(10 - xy^2), & 2 \leq x \leq 10, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the expected age of an insured automobile involved in an accident, $E[X]$.

4. **(12 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 2, x - 2 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[X^3Y].$$

5. **(12 points)** The profit of a new product is given by

$$Z = 3X - Y - 5.$$

Assume X and Y are independent random variables with

$$\text{Var}(X) = 1 \quad \text{and} \quad \text{Var}(Y) = 2.$$

Calculate $\text{Var}(Z)$.

6. **(12 points)** The profit of a new product is given by

$$Z = 5X - 2Y - 3.$$

Assume

$$\text{Var}(X) = 1, \quad \text{Var}(Y) = 2, \quad \text{Cov}(X, Y) = -1.$$

Calculate $\text{Var}(Z)$.

7. **(12 points)** An actuary analyzes a company's annual personal auto claim, M , and annual commercial auto claim, N . The analysis reveals that

$$\text{Var}(M) = 1600, \quad \text{Var}(N) = 900,$$

and the correlation between M and N is

$$\rho(M, N) = 0.64.$$

Calculate

$$\text{Var}(M + N).$$

8. **(12 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{8}{3}xy, & 0 \leq x \leq 1, \ x \leq y \leq 2x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\text{Cov}(X, Y).$$

1. (15 points) Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) (8 points) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(b) (7 points) Determine whether X and Y are independent.

a)
$$f(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^1 4xy dy$$

Treat x as constant:

$$f_X(x) = 4x \int_0^1 y dy = 4x \left[\frac{y^2}{2} \right]_0^1 = 4x \left(\frac{1}{2} \right) = 2x \quad \rightarrow \quad f_X(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Treat y as constant:

$$f_Y(y) = 4y \int_0^1 x dx = 4y \left[\frac{x^2}{2} \right]_0^1 = 4y \left(\frac{1}{2} \right) = 2y \quad \rightarrow \quad f_Y(y) = \begin{cases} 2y, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

b) Independent if $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for all x, y .

$$f_{X,Y}(x, y) = (2x)(2y) = 4xy$$

Therefore, X and Y are independent.

marginal
PDFs = joint
PDFs ✓

2. (13 points) Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5}(x^2 + y), & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[2X - Y + XY].$$

$$E[2X - Y + XY] = 2E[X] - E[Y] + E[XY]$$

① ② ③

$$\begin{aligned} \textcircled{1} E[X] &= \int_0^1 \int_0^1 x \frac{6}{5}(x^2 + y) dy dx = \frac{6}{5} \int_0^1 x \left[\int_0^1 (x^2 + y) dy \right] dx = \frac{6}{5} \int_0^1 \left(x^3 + \frac{x}{2} \right) dx \\ &= \left(\frac{6}{5} \right) \left(\frac{1}{4} + \frac{1}{4} \right) = \frac{6}{5} \cdot \frac{1}{2} \end{aligned}$$

$$E[X] = 3/5$$

$$\begin{aligned} \textcircled{2} E[Y] &= \int_0^1 \int_0^1 y \frac{6}{5}(x^2 + y) dx dy = \frac{6}{5} \int_0^1 y \left[\int_0^1 (x^2 + y) dx \right] dy = \frac{6}{5} \int_0^1 \left(\frac{y}{3} + y^2 \right) dy \\ &= \left(\frac{6}{5} \right) \left(\frac{1}{6} + \frac{1}{3} \right) = \frac{6}{5} \left(\frac{1}{2} \right) \end{aligned}$$

$$E[Y] = 3/5$$

$$\begin{aligned} \textcircled{3} E[XY] &= \int_0^1 \int_0^1 xy \frac{6}{5}(x^2 + y) dy dx = \frac{6}{5} \int_0^1 \int_0^1 (x^3 y + xy^2) dy dx \\ &= \frac{6}{5} \left[\left(\int_0^1 x^3 dx \right) \left(\int_0^1 y dy \right) + \left(\int_0^1 x dx \right) \left(\int_0^1 y^2 dy \right) \right] = \frac{6}{5} \left[\left(\frac{1}{4} \right) \left(\frac{1}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{1}{3} \right) \right] \\ &= \frac{6}{5} \left(\frac{1}{8} + \frac{1}{6} \right) = \frac{6}{5} \left(\frac{3}{24} + \frac{4}{24} \right) = \frac{6}{5} \cdot \frac{7}{24} = \frac{7}{20} \end{aligned}$$

$$E[XY] = \frac{7}{20}$$

$$E[2X - Y + XY] = 2E[X] - E[Y] + E[XY] = 2\left(\frac{3}{5}\right) - \frac{3}{5} + \frac{7}{20} = \frac{3}{5} + \frac{7}{20} = \frac{12}{20} + \frac{7}{20}$$

① ② ③

$$E[2X - Y + XY] = 19/20$$

3. (12 points) Let X represent the age of an insured automobile involved in an accident. Let Y represent the length of time the owner has insured the automobile at the time of the accident. Suppose X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{64}(10 - xy^2), & 2 \leq x \leq 10, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the expected age of an insured automobile involved in an accident, $E[X]$.

$$\begin{aligned} E[X] &= \int_2^{10} \int_0^1 x f_{X,Y}(x,y) dy dx = \int_2^{10} \int_0^1 x \cdot \frac{1}{64} (10 - xy^2) dy dx \\ &= \frac{1}{64} \int_2^{10} \int_0^1 (10x - x^2 y^2) dy dx = \frac{1}{64} \int_2^{10} \left(10x - \frac{x^2}{3} \right) dx = \frac{1}{64} \left[5x^2 - \frac{x^3}{9} \right]_2^{10} \\ &\quad \underbrace{10x \int_0^1 dy - x^2 \int_0^1 y^2 dy = 10x - x^2 \left(\frac{1}{3} \right)}_{\text{integration}} \\ &= \frac{1}{64} \left(\frac{3500}{9} - \frac{172}{9} \right) = \frac{1}{64} \left(\frac{3328}{9} \right) \end{aligned}$$

$x=10: 5(10)^2 - \frac{(10)^3}{9} = 500 - \frac{1000}{9} = \frac{3500}{9}$
 $x=2: 5(2)^2 - \frac{(2)^3}{9} = 20 - \frac{8}{9} = \frac{172}{9}$

$$E[X] = 52/9 \approx 5.78 \text{ years}$$

4. (12 points) Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 2, x-2 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[X^3 Y].$$

$$\begin{aligned} E[X^3 Y] &= \int_0^2 \int_{x-2}^x x^3 y \left(\frac{1}{4} \right) dy dx = \frac{1}{4} \int_0^2 x^3 \left[\frac{y^2}{2} \right]_{x-2}^x dx = \frac{1}{8} \int_0^2 x^3 (y)_{x-2}^x dx \\ &\quad \underbrace{\times 1/2}_{\text{integration}} \\ &= \frac{1}{8} \int_0^2 x^3 [x^2 - (x-2)^2] dx = \frac{1}{8} \int_0^2 x^3 (4x - 4) dx = \frac{1}{2} \int_0^2 x^4 - x^3 dx = \frac{1}{2} \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^2 \\ &\quad \underbrace{x^2 - 4x + 4}_{\text{expansion}} \quad \underbrace{\times 4}_{\text{integration}} \\ &= \frac{1}{2} \left(\frac{32}{5} - \frac{16}{4} \right) = \frac{1}{2} \left(\frac{12}{5} \right) \end{aligned}$$

$$E[X^3 Y] = 6/5$$

5. (12 points) The profit of a new product is given by

$$Z = 3X - Y - 5.$$

$$\text{Cov}(X, Y) = 0$$

Assume X and Y are independent random variables with

$$\text{Var}(X) = 1 \quad \text{and} \quad \text{Var}(Y) = 2.$$

Calculate $\text{Var}(Z)$.

$$\begin{aligned} \text{Var}(Z) &= \text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y) \\ &= 3^2(1) + (-1)^2(2) \end{aligned}$$

$$\text{Var}(Z) = 11$$

6. (12 points) The profit of a new product is given by

$$Z = 5X - 2Y - 3.$$

Assume

$$\text{Var}(X) = 1, \quad \text{Var}(Y) = 2, \quad \text{Cov}(X, Y) = -1.$$

Calculate $\text{Var}(Z)$.

$$\begin{aligned} \text{Var}(Z) &= \text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y) \\ &= 5^2(1) + (-2)^2(2) + 2(5)(-2)(-1) \\ &= 25 + 8 + 20 \end{aligned}$$

$$\text{Var}(Z) = 53$$

7. (12 points) An actuary analyzes a company's annual personal auto claim, M , and annual commercial auto claim, N . The analysis reveals that

$$\text{Var}(M) = 1600, \quad \text{Var}(N) = 900,$$

and the correlation between M and N is

$$\rho(M, N) = 0.64.$$

Calculate

$$\text{Var}(M + N).$$

$$\begin{aligned} \rho(M, N) &= \frac{\text{Cov}(M, N)}{\sqrt{\text{Var}(M)\text{Var}(N)}} \rightarrow \text{Cov}(M, N) = \rho(M, N) \sqrt{\text{Var}(M)\text{Var}(N)} = 0.64 \left(\sqrt{1600 \cdot 900} \right) \\ &= 0.64(40)(30) \\ \text{Cov}(M, N) &= 768 \end{aligned}$$

$$\text{Var}(M+N) = \text{Var}(M) + \text{Var}(N) + 2\text{Cov}(M, N) = 1600 + 900 + 2(768) = 2500 + 1536$$

$$\text{Var}(M+N) = 4036$$

8. (12 points) Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{8}{3}xy, & 0 \leq x \leq 1, x \leq y \leq 2x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\text{Cov}(X, Y).$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned} \textcircled{1} E[X] &= \int_0^1 \int_x^{2x} x \left(\frac{8}{3}xy \right) dy dx = \frac{8}{3} \int_0^1 x^2 \left[\frac{y^2}{2} \right]_x^{2x} dx = \frac{8}{3} \int_0^1 x^2 \left(\frac{4x^2 - x^2}{2} \right) dx \\ &= 4 \int_0^1 x^4 dx = 4 \left(\frac{1}{5} \right) \end{aligned}$$

$$E[X] = 4/5$$

$$\begin{aligned} \textcircled{2} E[Y] &= \int_0^1 \int_x^{2x} y \left(\frac{8}{3}xy \right) dy dx = \frac{8}{3} \int_0^1 x \left[\frac{y^3}{3} \right]_x^{2x} dx = \frac{8}{3} \int_0^1 x \left(\frac{8x^3 - x^3}{3} \right) dx \\ &= 56/9 \int_0^1 x^4 dx = 56/9 \left(\frac{1}{5} \right) \end{aligned}$$

$$E[Y] = 56/45$$

↓ continue

$$\textcircled{3} E[XY] = \int_0^1 \int_x^{2x} xy \left(\frac{8}{3}xy\right) dy dx = \frac{8}{3} \int_0^1 x^2 \left[\frac{y^3}{3}\right]_x^{2x} dx = \frac{8}{3} \int_0^1 x^2 \left(\frac{8x^3 - x^3}{3}\right) dx$$

$$= \frac{56}{9} \int_0^1 x^5 dx = \frac{56}{9} \left(\frac{1}{6}\right) = \frac{28}{27}$$

$$E[XY] = \frac{28}{27}$$

$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = \frac{28}{27} - \left(\frac{4}{5}\right)\left(\frac{56}{45}\right) = \frac{28}{27} - \frac{224}{225}$$

calculator...

$$\text{Cov}(X,Y) = \frac{28}{675} \approx 0.0415$$